



# Michael Boiman

Freelance Quality Engineer & AI Architect

Frankfurt am Main, Germany | Available for new projects

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## CAREER PROFILE

**Quality Engineer and AI Architect.** Quality engineering and AI architecture are not consecutive chapters here, they are two tracks running in parallel and feeding into each other.

In enterprise projects (**DB Vertrieb, DVAG, TÜV Süd**), quality monitoring systems took shape, AI-driven test automation was added, and legacy migrations were secured with 24/7 validation.

Today that means e-invoicing to EN 16931 in production and a network of agents consulting each other over the open MCP and A2A protocols.

I also pass this knowledge on, in developer and business workshops and in an invited talk at TU Darmstadt, translating hands-on AI for technical and non-technical audiences alike. I have introduced doctors, lawyers and financial advisers to working with AI in their own daily work and supported business teams across several projects.

### EN 16931

e-invoicing in production  
Peppol BIS 3.0 into SAP ByDesign

### MIT

open-source agent infrastructure  
github.com/bks-lab/open-bridge

### ISTQB®

Certified Tester  
test method to the international standard

## COMPETENCIES

|                       |                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality Engineering   | Quality Engineering, Specification & Verification Design, Playwright, Behave / pytest (BDD), Cucumber/Gauge, JMeter, Gatling                                                                              |
| AI & Agents           | Claude Code (Skills, Hooks, MCP), Agent Harness Engineering, LLMs (Claude, Gemini, GPT), Agent Orchestration (MCP & A2A), a2a-sdk & Agent Cards, Google ADK, LangChain                                    |
| Delivery & Automation | Python, TypeScript, FastAPI, JSON-RPC 2.0 & SSE, CI/CD, Docker, REST APIs, Agile (Scrum/Kanban)                                                                                                           |
| Platform & Monitoring | Peppol/E-Invoicing (EN 16931), Azure Functions, Azure DevOps, Azure Monitor & KQL, Cloudflare (Workers, Pages, R2), Microsoft 365 / Exchange Online, Grafana, Elasticsearch, Kibana, OpenTelemetry / OTLP |
| More                  | AWS, Google Cloud                                                                                                                                                                                         |

## PROFESSIONAL EXPERIENCE

### AI Test Automation Engineer · Building the AI Test Automation Process

since 09/2026

KION Group

**AI-driven test automation** for one of the group's customer portals: building the complete process, from deriving the test cases to a green CI pipeline.

#### Key Responsibilities

- Test orchestrator:** a chain of AI building blocks with a fixed log schema, its own model tier per block and usage tracking, packaged as a reusable command for the team
- Architecture rules for the test code base:** derived from a multi-stage analysis with several AI agents, independently cross-checked and documented
- Architecture check for generated test code:** detects violations of the team's style guide and fixes the minor ones on its own
- Calibrated comparison runs:** AI-generated E2E tests compared on cost, usage and conformance to the standard, across several model tiers

#### Tools & Technologies

- Testing: Playwright, TypeScript, Angular code base
- CI/CD & tooling: GitHub Actions, Sonar, Jira with Xray, Confluence, SAFe
- AI: Claude Code (several model tiers), GitHub Copilot CLI

### Senior Quality Engineer · Test Automation and Monitoring

09/2025-09/2026

TÜV Süd

**Quality Engineering** for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

#### Key Responsibilities

- BDD API Test Framework:** Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- E2E Test Automation:** Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- Performance & Monitoring:** Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring

- **Bug Analysis & Quality Intelligence:** Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting

#### Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLAlchemy, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, Azure Blob Storage, Poetry
- Agentic Coding: GitHub Copilot CLI

### LLM Infrastructure Architect & Automation Engineer

since 06/2025

BKS · AI Research & Development (internal)

**Development of LLM-orchestrated Enterprise Development Ecosystem:** Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

#### Key Responsibilities

- **Open-Source Agent Infrastructure:** open-bridge under the MIT licence, public at [github.com/bks-lab/open-bridge](https://github.com/bks-lab/open-bridge), with a live demo site. The generic CORE layer (agent runtime, MCP gateway, skill tree, promote flow with a content-leak gate in CI) grows out of my own instance and is published upward by scope, with its own test suite and continuous releases since June 2026.
- **Harness engineering:** what an agent achieves is decided mostly outside the model: context, tool selection, permissions, state across sessions, verification. That layer is measured here rather than estimated. What a session loads before it answers the first word carries a declared ceiling that a CI step checks in bytes; rules that rarely bite load on their trigger word instead of at start-up; the project registries sit in context as a card and are fetched one entry at a time; a further check reports what exists in the tree but is no longer reachable from any file a session loads, and attacks itself to prove it still catches things. The tools behind this are public and run in public CI
- **Data separation as architecture, not as agreement:** separate instances per client, a rule forbidding access between instances, and a fail-closed contract for materialising shared configuration (the sync aborts rather than writing on doubt). Protection against data leakage at three points: pre-commit hook, pre-push hook and three validators in CI. Plus an adversarial simulation that attacks the protection itself on a schedule, as its own CI job in a sealed sandbox
- **A2A Agent Network:** an a2a-sdk server with a local claude backend, running on a runtime adopted from open-bridge. Three agents (person, company, OSS project) consult each other on read-only edges, fronted by a stateless MCP-to-A2A gateway so MCP-only clients can reach them. Agent Cards at protocol version 1.0, JSON-RPC 2.0. Around 4,600 lines of Python with 98 automated test cases, built as a team effort (four further contributors)

#### Tools & Technologies

- LLM Orchestration: Claude (primary), Google Gemini, OpenAI GPT, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration; Cursor before that
- Multi-Agent: A2A Protocol 1.0, a2a-sdk, Agent Cards, MCP-to-A2A gateway, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Microsoft Graph, Exchange Online, Elasticsearch, Kibana, OpenTelemetry/OTLP
- Operations: Cloudflare (Workers, Pages, R2, DNS), Tailscale, launchd, restic/rclone
- Cloud: AWS, Google Cloud (alongside Azure)
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems; orchestrated repos span Python, TypeScript, Rust and Go

### AI-driven QA Environment for Energy Infrastructure

07/2024-01/2025

AkkuSwap Startup

**QA Leadership for EU-wide Battery Swap Infrastructure:** AI-driven simulation tool for battery swap station network with energy infrastructure integration

#### Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

#### Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Validated proof-of-concept with measurable reliability improvements
- Created testing framework for energy sector critical systems

### Technical Lead & AI Automation Architect

01/2024-04/2025

BKS on behalf of a client in the digital economy

**Technical lead & development:** Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

#### Focus areas

- **Workflow automation:** Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction:** Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration:** Connection to European e-invoicing network through a certified Peppol Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration:** Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.

Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Access-Point-API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

AI Engineering Lead & ML Solutions Architect

04/2023-04/2024

BKS on behalf of a publishing house

**Project management & technical lead:** Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation.

Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021-05/2025

DVAG

**Test management & coordination,** as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

**Result:** Real-time sprint-quality transparency via Grafana dashboards, quality-gated releases in the teams' CI/CD pipelines: data-driven release decisions replacing manual data gathering.

Tools

- IntelliJ / Visual Studio Code
- Java / Kotlin / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- CI/CD / GitHub (reusable workflows)
- Helm / Containers
- JIRA / Confluence
- Grafana
- Behaviour Driven Development
- Consumer-Driven Contracts (Pact.io)

Founder: Boiman.Solutions and QualityCluster

06/2021-08/2021

Boiman.Solutions

**Set up** his own consultancy Boiman.Solutions and the quality engineering community QualityCluster, a group of quality engineers working towards more quality through a higher degree of automation.

DevOps Quality Lead & Dashboard Architect

01/2017-05/2021

DB Vertrieb

**Test management, test coordination, test automation, agile (Kanban, Scrum, SAFe):** Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

Tools

- IntelliJ
- Java
- Cucumber / Cypress
- JMeter / Gatling
- CI/CD / GitLab CI / Jenkins
- Helm / Docker
- JIRA / Confluence
- Kibana / Grafana / Instana / Graylog
- AWS
- Behaviour Driven Development
- Consumer Driven Contracts (Spring Cloud Contract)
- Ionic (hybrid app)
- Selenium

Earlier positions

09/2015-01/2017     **Performance Engineering Lead & Test Architect**  
DB Systel

04/2015-09/2015     **Senior iOS Developer & Mobile Architect**  
Telekom

|                 |                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05/2009-03/2015 | <b>Quality Assurance Lead &amp; Test Automation Architect</b><br>Siemens                                                                                                             |
| 03/2009-05/2009 | <b>Performance Test Engineer &amp; Load Testing Specialist</b><br>ING-DIBA                                                                                                           |
| 01/2006-12/2008 | <b>Performance Test Engineer &amp; Quality Consultant</b><br>British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank |

## SELECTED PROJECTS

### E-Invoicing Automation Platform

Python · Azure Functions · Peppol · Peppol Access Point · SAP ByDesign

Production e-invoicing pipeline compliant with EU standard EN 16931.

- **Problem:** Inbound and outbound invoices in multiple formats (ZUGFeRD, XRechnung, UBL) required EN 16931-compliant processing.
- **Solution:** Peppol connection through a certified access point, hexagonal architecture, automatic transfer into the ERP.
- **Result:** Production system, invoice processing runs automated end-to-end into the ERP. Seven repositories, roughly 2,900 automated test cases, more test code than production code, an enforced coverage threshold per repository.

### Agent mesh and MCP-to-A2A gateway

Python · a2a-sdk · A2A Protocol 1.0 · Agent Cards · JSON-RPC 2.0

Three agents, one translator, one open protocol. The agent on my CV page is one of them.

- **Problem:** An agent that only holds its own knowledge guesses at everything outside it.
- **Solution:** Three agents, each with its own subject, consult each other over the open A2A protocol, read-only. In front of them a stateless gateway offering the same agents to MCP clients as three tools.
- **Result:** Publicly retrievable agent card, 98 automated test cases, generic framework open under MIT.

### AI-driven Test Automation

Python · OpenAI GPT-4 · Playwright · IDE Integration · Jira API

Guideline-based AI agent in the IDE that generates tests from Jira stories.

- **Problem:** Deriving test cases from a story is manual work, and a model without constraints invents steps and selectors the application does not have.
- **Solution:** The agent reads the story over the Jira API; the project's own conventions (page objects, naming rules, permitted selectors) sit beside it as a rule file and bound what it may produce.
- **Result:** The test comes out in the style of the existing Playwright project. Typing an empty file turns into reviewing a draft.

### Appointment enquiry over WhatsApp and phone

Python · FastAPI · WhatsApp Business Cloud API · Speech-to-Text · Text-to-Speech

Two channels on one domain core, WhatsApp and phone, reading free slots live from the booking portal.

- **Problem:** Calls were lost during work, and a language model must never invent an appointment.
- **Solution:** The model runs the conversation, the free slots come live from the portal API. It pencils an appointment in but never books it.
- **Result:** The WhatsApp channel has been in production at the client since 08/2026 with real end users and data residency in Germany; consent comes before the first substantive sentence.

### 24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java · Cucumber · Kibana · Graylog · Elasticsearch

Dual-run validation for safely migrating a mainframe price calculation.

- **Problem:** Millions of price combinations had to migrate without quality loss or data errors.
- **Solution:** 24/7 validation with randomly generated tests, live dashboard and full deviation traceability.
- **Result:** Safe migration without data loss, all combinations validated around the clock.

### Quality Dashboard · Real-time Overview

Grafana · Elasticsearch · Python · CI/CD · Playwright

Real-time quality overview for enterprise go/no-go decisions.

- **Problem:** Release decisions required hours of gathering data from the scattered test tools of several teams and environments.
- **Solution:** End-to-end pipeline with live dashboard, multi-source integration and automated PDF reports.
- **Result:** Data-driven release decisions in real time instead of manual collection.

### AI Context Orchestrator: Multi-Repo Development Platform

Python · YAML · Claude Code Plugins · Hooks API · GitHub Actions

Central control system for AI-assisted development across customer, partner and internal repos.

- **Problem:** Development across many repos and multiple customers required constant context switching and manual coordination.
- **Solution:** Orchestration hub with a three-way split registry, one skill tree for every tool and a hook-based permission system.
- **Result:** Workflows from incident response to meeting documentation largely automated.

TALKS & WORKSHOPS

|         |                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 04/2026 | <b>Guest Lecture: AI-Assisted Software Development in Practice</b><br>TU Darmstadt · Information Systems Group · Lab course “AI Startup: From Idea to Execution” |
| 06/2025 | <b>Workshop: Development with Generative Language Models</b><br>Developer Workshop · Agent-based Software Development                                            |
| 04/2025 | <b>Presentation: Efficient Documentation through Automation</b><br>Enterprise Presentation · AI-powered Documentation Workflows                                  |
| 2023    | <b>Applications of AI in Business Context</b><br>AI Workshop for Business Integration                                                                            |

EDUCATION

|           |                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------|
| 2000-2005 | <b>Diploma in Computer Science (UAS)</b><br>Cologne University of Applied Sciences, Gummersbach |
| 2005      | <b>ISTQB Certified Tester</b>                                                                   |
| 2007      | <b>Certified Dynatrace Diagnostics Basic Training</b>                                           |

LANGUAGES

|         |             |
|---------|-------------|
| German  | Native      |
| English | Fluent (C1) |